



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2024– 2025 (Odd Semester)

Degree, Semester & Branch : VII Semester B.E CSE
Course Code & Title : CCS346 & Exploratory Data Analysis
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit II / EDA USING PYTHON/Data Indexing and Selection
- **Course Outcome:** CO2
- **Topic Learning Outcome:** 2a
- **Activity Chosen:** Creating Categories
- **Justification:**
 - ❖ This activity promotes collaborative learning by encouraging students to engage in free writing and categorizing.
 - ❖ Sorting quotes and claims helps students identify underlying assumptions about data indexing and selection, fostering deeper understanding of the topic.
 - ❖ By organizing their thoughts visually, students can see connections between ideas, which enhances critical thinking and analysis.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
 - ❖ Upon completing the second unit, the instructor employed an engaging activity to assess the students understanding and encourage deeper reflection on the material. This method also helped the instructor gauge the student's comprehension of specific topics of data indexing and provided an opportunity to address any lingering doubts.
 - ❖ The class was divided into four groups, each assigned a unique role:
 - Group Free writing (10 minutes)
 - Divide students into small groups of 4-5.
 - Each group receives the prompt: "What is your position on the importance of data indexing and selection in EDA?"
 - Groups spend 10 minutes free writing their ideas, focusing on aspects such as the role of indexing in data exploration, improving analysis efficiency, and ensuring data integrity.
 - Example Position Statements such as
 - "In EDA, effective data indexing is essential to streamline data access and enhance the analytical process."
 - "Proper selection of relevant data is crucial in EDA to uncover meaningful insights and avoid biases."

- **Sorting Quotes/Claims**
 - The instructor hands out strips of paper with various quotes and claims related to data indexing and selection in EDA.
 - **Examples of Quotes:**
 - “Data indexing reduces the time required to access large datasets, making EDA more efficient.”
 - “In EDA, selecting the right subset of data can dramatically influence the conclusions drawn from the analysis.”
 - “Poor data selection can lead to skewed results, undermining the entire exploratory process.”
 - Groups read through the quotes and begin sorting them into categories based on common themes or assumptions they recognize, such as: Efficiency in Analysis, Data Integrity and insight Discovery.
- **Incorporating Position Statements**
 - After sorting the quotes, each group adds their previously drafted position statements to the relevant categories.
 - Groups explain their reasoning for placing their statements in specific categories, fostering discussion and deeper analysis.
 - For the Efficiency in Analysis category, a group might add:
 - “In EDA, effective data indexing is essential to streamline data access and enhance the analytical process.”
 - They could explain:
 - “This statement highlights how indexing can significantly reduce the time spent on data retrieval, thus improving overall analysis efficiency.”
- ❖ After group discussions, the entire class reconvened for a brief session where each group presented their insights. This allowed for a comprehensive review of the material and ensured that all doubts were clarified.

❖ **CO – PO / PSO mapping:**

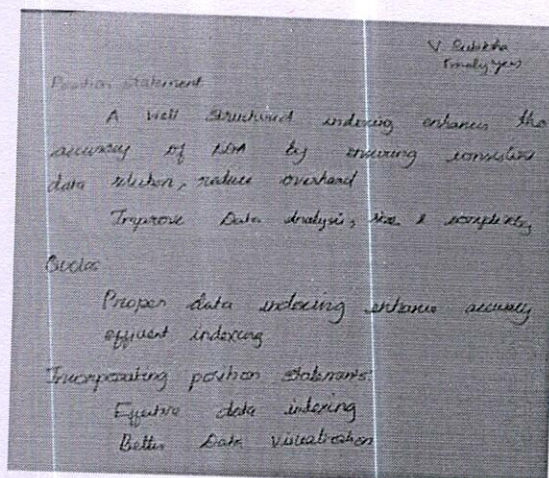
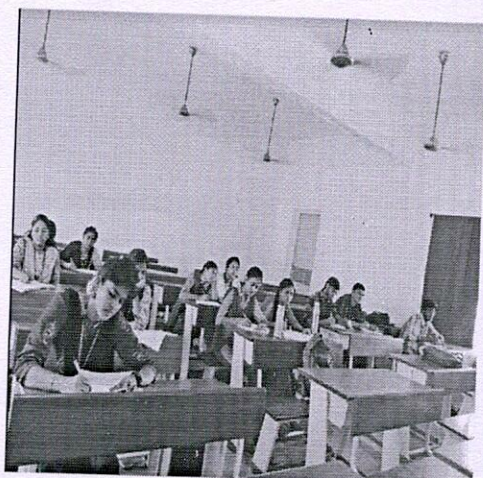
CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
CO1	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO9		PO10	PO12
	1		2	2
Justification correlation for	Student help in breaking down complex concepts into manageable parts, making it easier to communicate and collaborate within a team.		Written communication/ Asking questions are improved	Student will implement the questioning skill in future in written format

• Images / Screenshot of the practice:



Reflective Critique:

Conclude the activity with a class discussion, encouraging groups to share their insights and any surprising discoveries about the assumptions underlying their perspectives on data indexing and selection.

❖ **Discussing with students:**

- Students expressed appreciation for the collaborative aspect, noting that discussing ideas in groups helped clarify their understanding.
- The sorting process revealed diverse perspectives and encouraged critical dialogue about the assumptions underpinning data indexing.

❖ **Benefit of the practice:**

- ☐ Improved clarity on the topic of data indexing and selection.
- ☐ Enhanced ability to articulate positions based on evidence and shared assumptions.

❖ **Challenges faced in implementation:**

- Some groups needed additional guidance on how to categorize quotes effectively and may have benefitted from more examples.
- Time constraints made it difficult for some groups to complete the sorting process thoroughly.

References:

- <https://teaching.unl.edu/resources/lecture-reaction/>
- <https://www.skillshub.com/blog/active-learning-strategies-examples/>
- <https://www.prodigygame.com/main-en/blog/active-learning-strategies-examples/>

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